Yinoussa Adagolodjo | Assistant Professor in Computer Science

DEFROST Team, Inria · University of Lille – Villeneuve-d'Ascq, France
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adagolodjo
Google Scholar | ResearchGate | HAL Science | ORCID

Research Interests

Primary: Medical Robotics, Real-time Finite Element Simulation, Computer-Assisted Surgery

Secondary: Soft Robotics, Augmented Reality, Constraint-Based Control, Haptic Rendering

Applications: Robotic needle insertion · Cochlear implant robotization · Surgical AR · Continuum robot control

Professional Experience

University of Lille · DEFROST Team, Inria — **Lille, France**
Assistant Professor in Computer Science — *2022–present*

- Research in medical robotics and real-time simulation for healthcare
- Supervision of PhD students and postdoctoral researchers
- Teaching at Polytech Lille (Industrial Engineering, Biological & Food Engineering, PeiP) and in the Master in Automatic Control & Robotics: medical robotics, automation, programming, mathematics, Industry 4.0
- PI of ACCESS (follow-up of ROBOCOP) · member of the IRE (European) research project

Inria · DEFROST Team — **Lille, France**
Research Engineer — *2019–2022*

- Constraint-based control for soft and hybrid robotic systems (ROBOCOP, COSSEROOTS)
- Real-time FEM simulation for robotic cochlear implant insertion
- Lead developer of the Cosserat plugin for SOFA (open source, C++ + Python)

ICube Laboratory · University of Strasbourg — **Strasbourg, France**
Doctoral Researcher — *2015–2018*

- PhD: robotic needle insertion using inverse FEM simulation (CONECT project)
- Augmented reality for open liver surgery and deformable organ registration
- Advisors: Dr. Hadrien Courtecuisse, Prof. Michel de Mathelin

Education

University of Strasbourg — **Strasbourg, France**
Ph.D. in Robotics & Computer Science — *2018*
Thesis: *"CONECT — Couplage de la rObotique et de la simulatioN médicalE pour des proCédures automaTisées"*
Advisors: Dr. H. Courtecuisse · Prof. M. de Mathelin

Involved Research Projects

National project · Team Member
ADAGIO — Continuum Robots for Endoscopic Gallbladder Drainage — *2026–present*
tendon-Actuated continuum robots for enDoscopic drAinage of the Gallbladder In acute chOlecystitis. Developing a stabilization method for stent placement through the precise control of continuum robots.

ANR ADAGIO · PI
Maestro — Advanced Endoscope for Robot-Assisted SpiderMass Imaging — *2024–present*
Design, modeling, and control of advanced endoscopes for in vivo real-time topography mass spectrometry imaging (SpiderMass) in robot-assisted minimally invasive surgery.

European Project · Team Member
IRE — Intelligent Robotic Endoscopes — *2024–present*
Soft robotic endoscopy with real-time AR guidance, computer vision, and ML for minimally invasive surgery.

Regional PhD grant (Hauts-de-France) · PI
Active Prostate Phantoms — DRE Training & Needle-Insertion Robotics *2023–present*
PhD thesis of Sizhe Tian. Sensorized, pneumatically actuated prostate phantoms with two goals: (1) teaching the digital rectal examination (DRE) to urology trainees; (2) improving the algorithms of needle-insertion robots.

Research Project · PI
TAVI Growing Robot — Transcatheter Aortic Valve Implantation *2023–present*
Vine-inspired eversion growing robots for Transcatheter Aortic Valve Implantation (TAVI): design, modeling, and control for cardiovascular navigation in constrained anatomical spaces.

Follow-up of ROBOCOP · PI
ACCESS — Digital Twin for Cochlear Implant Surgery *2024–present*
Patient-specific digital twin for cochlear implant surgery: physical simulation, medical image processing, and augmented reality for intraoperative surgical guidance.

Inria · PI since 2022
ROBOCOP — Robotic Cochlear Implant Procedure *2019–2023*
Automated cochlear implant insertion via real-time FEM simulation, surgical navigation, and safety protocols.

European project · Invited partner
CoBra — Robotic Assistance for Transrectal Prostate Biopsy
Planning of a soft-rigid hybrid arm in contact with compliant tissue for transrectal prostate biopsy (IEEE RA-L 2022).

Postdoctoral Researcher
COSSEROOTS — Cosserat Theory for Soft Robotics Control *2021–2022*
Advanced control strategies for continuum soft robots using Cosserat rod theory.

Doctoral Researcher
CONECT — Control of Robotic Needle Insertion *2015–2018*
Real-time inverse FEM simulation for robotic needle control in deformable tissue with AR guidance.

Publications

Full list with citation metrics: HAL · Google Scholar · ResearchGate

Journal Articles — Under Review

2026: **A Sliding Force Formulation for Contact Localization: Application to a Bio-inspired Active Prostate Phantom for DRE Training**
S. Tian, **Y. Adagolodjo**, T. Navez, C. Duriez, J. Dequidt
IEEE Transactions on Robotics — major revision, resubmitted.

2026: **A Thin-Shell FEM Model of Soft Everting Robot Eversion: Material Comparison, Anatomical Validation and Preclinical Feasibility for TAVI**
F. Przybylski, **Y. Adagolodjo**, C. Saija, J. Dequidt, C. Duriez, P. Berthet-Rayne
IEEE/ASME Transactions on Mechatronics — major revision, resubmitted.

Journal Articles

2022: **Planning of Soft-Rigid Hybrid Arms in Contact with Compliant Environment: Application to the Transrectal Biopsy of the Prostate**
E. Coevoet, **Y. Adagolodjo**, M. Lin, C. Duriez, F. Ficuciello
IEEE Robotics and Automation Letters, 7(2), pp. 4853–4860. DOI · HAL

2021: **Coupling Numerical Deformable Models in Global and Reduced Coordinates for the Simulation of the Direct and the Inverse Kinematics of Soft Robots**
Y. Adagolodjo, F. Renda, C. Duriez
IEEE Robotics and Automation Letters, 6(2), pp. 3910–3917. DOI · HAL

2019: **Robotic Insertion of Flexible Needle in Deformable Structures Using Inverse Finite-Element Simulation**
Y. Adagolodjo, L. Goffin, M. de Mathelin, H. Courtecuisse
IEEE Transactions on Robotics, 35(3), pp. 697–708. DOI · HAL

Conference Papers

2026: **Compliance-Based Sensor Placement for Force Sensing on a Sensorized Prostate Phantom**
S. Tian, **Y. Adagolodjo**, J. Dequidt
CRAS 2026 — Conf. on New Technologies for Computer and Robot Assisted Surgery (accepted). HAL · arXiv

2025: **Active Prostate Phantom with Multiple Chambers**
S. Tian, **Y. Adagolodjo**, J. Dequidt

IROS 2025 — IEEE/RSJ Int. Conf. on Intelligent Robots and Systems. DOI · HAL

2025: **In silicone and in silico: toward evaluation of pacemaker lead implantation based on soft robotics and computer simulation**
T. Moupfouma, Q. Peyron, **Y. Adagolodjo**, S. Caubet, J.-F. Ollivier, C. Duriez
RoboSoft 2025 — IEEE Int. Conf. on Soft Robotics. DOI · HAL

2024: **Enhancing TAVI Robot Flexibility Using Semi-Extensible Materials**
F. Przybylski, **Y. Adagolodjo**, J. Dequidt, C. Duriez, P. Berthet-Rayne
HSMR 2024 — Hamlyn Symposium on Medical Robotics. DOI · HAL

2023: **3D Kinematics and Quasi-Statics of a Growing Robot Eversion**
F. Przybylski, **Y. Adagolodjo**, A. Mîra, G. Cerruti, J. Dequidt, C. Duriez, P. Berthet-Rayne
RoboSoft 2023 — IEEE Int. Conf. on Soft Robotics. DOI · HAL

2023: **Smart electrode array for cochlear implants**
A. Itawi, S. Ghenna, . . . , **Y. Adagolodjo**, . . . , C. Duriez, E. Cattan
IEEE MEMS 2023 — Int. Conf. on Micro Electro Mechanical Systems. DOI · HAL

2018: **Marker-based Registration for Large Deformations — Application to Open Liver Surgery**
Y. Adagolodjo, N. Golse, E. Vibert, M. de Mathelin, S. Cotin, H. Courtecuisse
ICRA 2018 — IEEE Int. Conf. on Robotics and Automation. DOI · HAL

2017: **Silhouette-based Pose Estimation for Deformable Organs — Application to Surgical Augmented Reality**
Y. Adagolodjo, R. Trivisonne, N. Haouchine, S. Cotin, H. Courtecuisse
IROS 2017 — IEEE/RSJ Int. Conf. on Intelligent Robots and Systems. DOI · HAL

2016: **Inverse real-time Finite Element simulation for robotic control of flexible needle insertion in deformable tissues**
Y. Adagolodjo, L. Goffin, M. de Mathelin, H. Courtecuisse
IROS 2016 — IEEE/RSJ Int. Conf. on Intelligent Robots and Systems. DOI · HAL

2015: **Haptic Rendering of Hyperelastic Models with Friction**
H. Courtecuisse, **Y. Adagolodjo**, H. Delingette, C. Duriez
IROS 2015 — IEEE/RSJ Int. Conf. on Intelligent Robots and Systems. DOI · HAL

Workshops & Posters

2024: **Using Haptic Feedback in Digital Rectal Examination Training**
S. Tian, **Y. Adagolodjo**, J. Dequidt
Workshop on Haptics and Simulation. DOI · HAL

2022: **How electroactive polymer can be used for cochlear implants micro-robotization?**
A. Itawi, S. Ghenna, . . . , **Y. Adagolodjo**, . . . , C. Duriez, E. Cattan
Int. Workshop on Biomaterial Innovations Toward Repair of the Human Body (poster). HAL

2015: **Domain Decomposition for real time Simulation of needle insertion**
Y. Adagolodjo, H. Courtecuisse, R. Trivisonne, L. Goffin, S. P.-A. Bordas, M. de Mathelin
DD23 — Int. Conf. on Domain Decomposition Methods (poster). HAL

PhD Thesis

2018: **CONECT — Couplage de la rObotique et de la simulatioN médicalE pour des proCédures automaTisées**
Ph.D. Thesis, University of Strasbourg. HAL tel-01939268

Invited Talks & Keynotes

University of Leeds, UK
COSUR 2026 — Summer School on Control of Surgical Robots (Keynote) *Sep. 2026*
"What Does a Surgical Robot Really Need to Know About the Human Body?"
Pre-event of CRAS 2026.

Lisbon, Portugal
COSUR 2025 — Summer School on Control of Surgical Robots *Sep. 2025*
"Modeling and Control of Deformable Robots Using the Finite Element Method"
Pre-event of CRAS 2025.

Santiago, Chile
Journées Scientifiques Inria Chile 2024 *Dec. 2024*
"Bridging the Gap: From Biomechanical Simulations to Adaptive Surgical Robotics"
Official pre-event of the AI Action Summit.

Rancagua, Chile
LACORO 2024 — 9th Latin American Conference on Robotics *Dec. 2024*
"From biomechanical simulation to deformable robot control"
Hong Kong
CAIR Hong Kong AI and Robotics in Healthcare Forum *Mar. 2024*
"Exploring the Frontiers of Deformable Robotics: Physics-based Modeling, Simulation and Control"
Centre for Artificial Intelligence and Robotics, Chinese Academy of Sciences.

Teaching Experience

Programs: Polytech Lille: Industrial Engineering (GI), Biological & Food Engineering (GBA), PeiP preparatory cycle · Master in Automatic Control & Robotics

Areas: Medical Robotics & Simulation · Automatic Control · Industrial Logic · Industrial Robotics (ABB) · Programming · Mathematics & Numerical Analysis · Industry 4.0 (competency-based) · Research Methods

Website: adagolodjo.com/teaching

Open Source Software

Cosserat Plugin for SOFA Framework **Open Source · C++ / Python**
Lead Developer *2019–present*
Real-time Cosserat rod mechanics for soft and continuum robot simulation. Internationally adopted by the soft robotics community.
`github.com/SofaDefrost/Cosserat`

Technical Skills

Programming: C++, Python, MATLAB, JavaScript, LaTeX

Frameworks: SOFA (real-time FEM) · ROS · OpenGL / WebGL

Methods: Finite Element Method · Cosserat rod theory · GPU computing · Constraint-based control

Tools: Git · Docker · Jekyll · Linux / macOS

Languages

French: Native speaker

English: Fluent *Professional working proficiency*

Yoruba: Native speaker

Bassar: Native speaker

Professional Service

Journals: Reviewer for *IEEE Transactions on Robotics* · *IEEE Trans. on Medical Robotics and Bionics* · *IEEE Robotics and Automation Letters*

Conferences: Reviewer for ICRA · IROS · Int. Conf. on Medical Robotics

Membership: IEEE Robotics and Automation Society